Rank 2 packages independent moment directions

Moment matrix

$$\begin{bmatrix} \mathrm{Li}_1(t) & \mathrm{Li}_2(t) \\ \mathrm{Li}_2(t) & \mathrm{Li}_3(t) \end{bmatrix}$$

$$H_2(t) = \mathrm{Li}_1(t)\,\mathrm{Li}_3(t) - \mathrm{Li}_2(t)^2$$

Each pair $1 \le i < j$ contributes

$$t^{(i+j)} \cdot (j-i)^2 / (i^3\, j^3)$$

The first pair is $\{1,2\}$: $H_2(t) = t^3/8 + \cdots$